

Bivariate normal Distribution

①

Let the Two Dimensional random Variable $(X; Y)$ have the joint probability density function.

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}\left\{\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) + \left(\frac{y-\mu_2}{\sigma_2}\right)^2\right\}\right]$$

$$\begin{aligned} -\infty < x < \infty & \quad -\infty < \mu_1 < \infty \\ -\infty < y < \infty & \quad -\infty < \mu_2 < \infty \\ -1 < \rho < 1 & \quad \sigma_1, \sigma_2 > 0 \end{aligned}$$

Question

of

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}\left\{\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) + \left(\frac{y-\mu_2}{\sigma_2}\right)^2\right\}\right]$$

Show that $f(x, y)$ is a probability density function.

Sol A function $f(x, y)$ is a probability density function

$$\text{if } \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) = 1$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}\left\{\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) + \left(\frac{y-\mu_2}{\sigma_2}\right)^2\right\}\right] dx dy$$

$$\begin{aligned} \text{put } \frac{x-\mu_1}{\sigma_1} &= u & \frac{y-\mu_2}{\sigma_2} &= v \\ x &= \mu_1 + \sigma_1 u & y &= \mu_2 + \sigma_2 v \\ dx &= \sigma_1 du & dy &= \sigma_2 dv \end{aligned}$$

$$\frac{\partial x}{\partial u} = \frac{dx}{du} = \sigma_1 \quad \frac{\partial y}{\partial v} = \frac{dy}{dv} = \sigma_2$$

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{vmatrix} \Rightarrow \sigma_1 \sigma_2 \quad (2)$$

$$dy dx = J dv du$$

$$= \sigma_1 \sigma_2 dv du$$

$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2(1-\rho^2)}\{u^2 - 2\rho uv + v^2\}\right] \sigma_1 \sigma_2 dv du$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2(1-\rho^2)}\{u^2 - 2\rho uv + \rho^2 v^2 - \rho^2 v^2 + v^2\}\right] dv du$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2(1-\rho^2)}\{(u-\rho v)^2 + v^2(1-\rho^2)\}\right] dv du$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left[-\frac{(u-\rho v)^2}{2(1-\rho^2)} - \frac{v^2(1-\rho^2)}{2(1-\rho^2)}\right] dv du$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}\left(\frac{u-\rho v}{\sqrt{1-\rho^2}}\right)^2 - \frac{1}{2}v^2\right] dv du$$

put $\frac{u-\rho v}{\sqrt{1-\rho^2}} = w$

$$u - \rho v = \sqrt{1-\rho^2} w$$

$$u = \rho v + \sqrt{1-\rho^2} w$$

$$du = \sqrt{1-\rho^2} dw$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}w^2 - \frac{1}{2}v^2\right] dv \sqrt{1-\rho^2} dw$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}w^2 - \frac{1}{2}v^2\right] dv dw$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{W^2}{2}} dW \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{V^2}{2}} dV$$

put $\frac{W^2}{2} = m$

$$W^2 = 2m$$

$$W dW = dm$$

$$dW = \frac{dm}{W}$$

$$= \frac{dm}{\sqrt{2m}}$$

$$\frac{V^2}{2} = t$$

$$V^2 = 2t$$

$$V dV = dt$$

$$dV = \frac{dt}{V}$$

$$= \frac{dt}{\sqrt{2t}}$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-m} \frac{dm}{\sqrt{2m}} \cdot \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-t} \frac{dt}{\sqrt{2t}}$$

$$= \frac{1}{\sqrt{\pi}} \int_0^{\infty} e^{-m} m^{-\frac{1}{2}} dm \cdot \frac{1}{\sqrt{\pi}} \int_0^{\infty} e^{-t} t^{-\frac{1}{2}} dt$$

$$= \frac{1}{\sqrt{\pi}} \int_0^{\infty} e^{-m} m^{\frac{1}{2}-1} dm \cdot \frac{1}{\sqrt{\pi}} \int_0^{\infty} e^{-t} t^{\frac{1}{2}-1} dt$$

$$= \frac{1}{\sqrt{\pi}} \cdot \sqrt{\frac{1}{2}} \cdot \frac{1}{\sqrt{\pi}} \cdot \sqrt{\frac{1}{2}}$$

$$= \frac{1}{\sqrt{\pi}} \cdot \sqrt{\pi} \cdot \frac{1}{\sqrt{\pi}} \cdot \sqrt{\pi}$$

$$= 1 \cdot 1 = 1$$

Question prove That the m.g.f. of the Bivariate Normal Distribution is. (4)

$$M_{X,Y}(t_1, t_2) = e^{\mu_1 t_1 + \mu_2 t_2 + \frac{1}{2}(\sigma_1^2 t_1^2 + 2\rho\sigma_1\sigma_2 t_1 t_2 + \sigma_2^2 t_2^2)}$$

Sol By definition

$$M_{X,Y}(t_1, t_2) = E(e^{t_1 x + t_2 y}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{t_1 x + t_2 y} f(x, y) dy dx$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{t_1 x + t_2 y} \exp\left[-\frac{1}{2(1-\rho^2)}\left\{\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) + \left(\frac{y-\mu_2}{\sigma_2}\right)^2\right\}\right] dy dx$$

put $\frac{x-\mu_1}{\sigma_1} = u$

$$x = \mu_1 + \sigma_1 u$$

$$dx = \sigma_1 du$$

$$\frac{\partial x}{\partial u} = \frac{dx}{du} = \sigma_1$$

$\frac{y-\mu_2}{\sigma_2} = v$

$$y = \mu_2 + \sigma_2 v$$

$$dy = \sigma_2 dv$$

$$\frac{\partial y}{\partial v} = \frac{dy}{dv} = \sigma_2$$

$$|J| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{vmatrix} = \sigma_1 \sigma_2$$

$$dy dx = |J| dv du = \sigma_1 \sigma_2 dv du$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{t_1(\mu_1 + \sigma_1 u) + t_2(\mu_2 + \sigma_2 v)} \exp\left[-\frac{1}{2(1-\rho^2)}\{u^2 - 2\rho uv + v^2\}\right] \sigma_1 \sigma_2 dv du$$

$$= \frac{e^{\mu_1 t_1 + \mu_2 t_2}}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{\sigma_1 t_1 u + \sigma_2 t_2 v} \exp\left[-\frac{1}{2(1-\rho^2)}\{u^2 - 2\rho uv + v^2\}\right] dv du$$

$$= \frac{e^{\mu_1 t_1 + \mu_2 t_2}}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2(1-\rho^2)}\{u^2 - 2\rho uv + v^2\} + \sigma_1 t_1 u + \sigma_2 t_2 v\right] dv du$$

$$= \frac{\mu_1 t_1 + \mu_2 t_2}{2\pi \sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2(1-\rho^2)} \left\{ u^2 - 2\rho uv + v^2 - 2(1-\rho^2)t_1\sigma_1 u - 2(1-\rho^2)\sigma_2 t_2 v \right\} \right] du dv \quad (5)$$

$$\exp \left[-\frac{1}{2(1-\rho^2)} \left\{ u^2 - 2\rho uv + v^2 - 2(1-\rho^2)t_1\sigma_1 u - 2(1-\rho^2)\sigma_2 t_2 v \right\} \right] \quad (1)$$

$$\exp \left[-\frac{1}{2(1-\rho^2)} \left\{ u^2 - 2\rho uv - 2(1-\rho^2)t_1\sigma_1 u + v^2 - 2(1-\rho^2)\sigma_2 t_2 v \right\} \right]$$

$$\exp \left[-\frac{1}{2(1-\rho^2)} \left\{ u^2 - 2u(\rho v + (1-\rho^2)t_1\sigma_1) + (\rho v + (1-\rho^2)t_1\sigma_1)^2 - (\rho v + (1-\rho^2)t_1\sigma_1)^2 + v^2 - 2(1-\rho^2)\sigma_2 t_2 v \right\} \right]$$

$$\exp \left[-\frac{1}{2(1-\rho^2)} \left\{ (u - (\rho v + (1-\rho^2)t_1\sigma_1))^2 - \rho^2 v^2 - (1-\rho^2)t_1^2\sigma_1^2 - 2\rho v(1-\rho^2)t_1\sigma_1 + v^2 - 2(1-\rho^2)\sigma_2 t_2 v \right\} \right]$$

$$\exp \left[-\frac{1}{2(1-\rho^2)} \left\{ (u - (\rho v + (1-\rho^2)t_1\sigma_1))^2 + v^2 - \rho^2 v^2 - 2\rho v(1-\rho^2)t_1\sigma_1 - 2(1-\rho^2)\sigma_2 t_2 v - (1-\rho^2)t_1^2\sigma_1^2 \right\} \right]$$

$$\exp \left[-\frac{1}{2(1-\rho^2)} \left\{ (u - (\rho v + (1-\rho^2)t_1\sigma_1))^2 + v^2(1-\rho^2) - 2v(1-\rho^2)\rho t_1\sigma_1 - 2(1-\rho^2)\sigma_2 t_2 v - (1-\rho^2)t_1^2\sigma_1^2 \right\} \right]$$

$$\exp \left[-\frac{1}{2(1-\rho^2)} \left\{ (u - (\rho v + (1-\rho^2)t_1\sigma_1))^2 + (1-\rho^2)(v^2 - 2v(\rho t_1\sigma_1 + \sigma_2 t_2) - (1-\rho^2)t_1^2\sigma_1^2) \right\} \right]$$

$$\exp \left[-\frac{1}{2(1-\rho^2)} \left\{ (u - (\rho v + (1-\rho^2)t_1\sigma_1))^2 + (1-\rho^2)[v^2 - 2v(\rho t_1\sigma_1 + \sigma_2 t_2) + (\rho t_1\sigma_1 + \sigma_2 t_2)^2 - (\rho t_1\sigma_1 + \sigma_2 t_2)^2 - (1-\rho^2)t_1^2\sigma_1^2] \right\} \right]$$

$$\exp\left[-\frac{1}{2(1-\rho^2)}\left\{(u - \rho v - (1-\rho^2)t_1\sigma_1)^2 + (1-\rho^2)\left[V - (\rho t_1\sigma_1 + t_2\sigma_2)\right]^2 - (1-\rho^2)(\rho t_1\sigma_1 + \sigma_2 t_2)^2 - (1-\rho^2)\sigma_1^2 t_1^2\right\}\right] \quad (6)$$

$$\exp\left[-\frac{1}{2(1-\rho^2)}\left\{(u - \rho v - (1-\rho^2)\sigma_1 t_1)^2 + (1-\rho^2)\left[V - (\rho t_1\sigma_1 + t_2\sigma_2)\right]^2 - (1-\rho^2)\left[(\rho t_1\sigma_1 + \sigma_2 t_2)^2 + (1-\rho^2)\sigma_1^2 t_1^2\right]\right\}\right]$$

$$\exp\left[-\frac{1}{2(1-\rho^2)}\left\{(u - \rho v - (1-\rho^2)\sigma_1 t_1)^2 + (1-\rho^2)\left[V - \rho t_1\sigma_1 - t_2\sigma_2\right]^2 - (1-\rho^2)\left[\rho^2 t_1^2 \sigma_1^2 + \sigma_2^2 t_2^2 + 2\rho t_1 t_2 \sigma_1 \sigma_2 + \sigma_1^2 t_1^2 - \rho^2 t_1^2\right]\right\}\right]$$

$$\exp\left[-\frac{1}{2(1-\rho^2)}\left\{(u - \rho v - (1-\rho^2)\sigma_1 t_1)^2 + (1-\rho^2)\left[V - \rho t_1\sigma_1 - t_2\sigma_2\right]^2 - (1-\rho^2)\left[\sigma_1^2 t_1^2 + \sigma_2^2 t_2^2 + 2\rho t_1 t_2 \sigma_1 \sigma_2\right]\right\}\right]$$

$$\exp\left[-\frac{1}{2} \frac{(u - \rho v - (1-\rho^2)\sigma_1 t_1)^2}{(1-\rho^2)} - \frac{1}{2} (V - \rho t_1\sigma_1 - t_2\sigma_2)^2 + \frac{1}{2} (\sigma_1^2 t_1^2 + \sigma_2^2 t_2^2 + 2\rho\sigma_1\sigma_2 t_1 t_2)\right]$$

put in (4)

$$M_{(X,Y)}(t_1, t_2) = \frac{e^{u_1 t_1 + u_2 t_2}}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2} \frac{(u - \rho v - (1-\rho^2)\sigma_1 t_1)^2}{(1-\rho^2)} - \frac{1}{2} (V - \rho t_1\sigma_1 - t_2\sigma_2)^2 + \frac{1}{2} (\sigma_1^2 t_1^2 + \sigma_2^2 t_2^2 + 2\rho\sigma_1\sigma_2 t_1 t_2)\right] dv du$$

$$= e^{u_1 t_1 + u_2 t_2 + \frac{1}{2}(\sigma_1^2 t_1^2 + \sigma_2^2 t_2^2 + 2\rho\sigma_1\sigma_2 t_1 t_2)} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2} \frac{(u - \rho v - (1-\rho^2)\sigma_1 t_1)^2}{(1-\rho^2)} - \frac{1}{2} (V - \rho t_1\sigma_1 - t_2\sigma_2)^2\right] dv du$$

⑦

$$\text{put } \frac{u - pV - (1-p^2)\sigma_1 t_1}{\sqrt{1-p^2}} = W$$

$$u = pV + (1-p^2)\sigma_1 t_1 + \sqrt{1-p^2} W$$

$$du = \sqrt{1-p^2} dW$$

$$\frac{\partial u}{\partial W} = \sqrt{1-p^2}$$

$$V - p t_1 \sigma_1 - t_2 \sigma_2 = z$$

$$V = p t_1 \sigma_1 + t_2 \sigma_2 + z$$

$$dV = dz$$

$$\frac{dV}{dz} = \frac{\partial V}{\partial z} = 1$$

$$|J| = \begin{vmatrix} \frac{\partial u}{\partial W} & \frac{\partial u}{\partial z} \\ \frac{\partial V}{\partial W} & \frac{\partial V}{\partial z} \end{vmatrix}$$

$$= \begin{vmatrix} \sqrt{1-p^2} & 0 \\ 0 & 1 \end{vmatrix} = \sqrt{1-p^2}$$

$$dudV = |J| dW dz$$

$$= \sqrt{1-p^2} dW dz$$

$$M_{(x,y)}^{(t_1,t_2)} = \frac{e^{\mu_1 t_1 + \mu_2 t_2 + \frac{1}{2}(\sigma_1^2 t_1^2 + \sigma_2^2 t_2^2 + 2\rho\sigma_1\sigma_2 t_1 t_2)}}{2\pi \sqrt{1-p^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{e^{-\frac{1}{2}W^2 - \frac{1}{2}z^2}}{\sqrt{1-p^2}} dW dz$$

$$= e^{\mu_1 t_1 + \mu_2 t_2 + \frac{1}{2}(\sigma_1^2 t_1^2 + \sigma_2^2 t_2^2 + 2\rho\sigma_1\sigma_2 t_1 t_2)} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}W^2} dW \cdot \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz$$

$$M_{(x,y)}^{(t_1,t_2)} = e^{\mu_1 t_1 + \mu_2 t_2 + \frac{1}{2}(\sigma_1^2 t_1^2 + \sigma_2^2 t_2^2 + 2\rho\sigma_1\sigma_2 t_1 t_2)} \cdot 1$$

exm If (X, Y) has bivariate Normal distribution⁽²⁾

then show that

$$E(X) = \mu_1, \quad \text{Var}(X) = \sigma_1^2$$

$$E(Y) = \mu_2, \quad \text{Var}(Y) = \sigma_2^2$$

$$\text{Cor}(X, Y) = \rho \sigma_1 \sigma_2$$

$$\rho_{xy} = \rho$$

Question

If (X, Y) has a Bivariate Normal distribution, then the marginal distribution of X and Y are univariate i.e. X is normally distributed with mean μ_1 and Variance σ_1^2 or $X \sim N(\mu_1, \sigma_1^2)$ and Y is normally distributed with mean μ_2 and Variance σ_2^2 or $Y \sim N(\mu_2, \sigma_2^2)$

Sol The marginal density function of the Variable X is

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) + \left(\frac{y-\mu_2}{\sigma_2}\right)^2 \right\}\right] dy$$

$$\text{put } \frac{y-\mu_2}{\sigma_2} = V \quad y = \mu_2 + \sigma_2 V$$

$$dy = \sigma_2 dV$$

$$g(x) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)V + V^2 \right\}\right] \sigma_2 dV$$

Adding and Subtracting $\rho^2\left(\frac{x-\mu_1}{\sigma_1}\right)^2$

$$g(x) = \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2(1-\rho^2)}\left\{\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)V + V^2 + \rho^2\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - \rho^2\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right\}\right] dV \quad (7)$$

$$= \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2(1-\rho^2)}\left\{\left(\frac{x-\mu_1}{\sigma_1}\right)^2(1-\rho^2) + \left(V - \rho\left(\frac{x-\mu_1}{\sigma_1}\right)\right)^2\right\}\right] dV$$

$$= \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - \frac{1}{2}\left(\frac{V - \rho\left(\frac{x-\mu_1}{\sigma_1}\right)}{\sqrt{1-\rho^2}}\right)^2\right] dV$$

$$= \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - \frac{1}{2}\left(\frac{V - \rho\left(\frac{x-\mu_1}{\sigma_1}\right)}{\sqrt{1-\rho^2}}\right)^2\right] dV$$

put $\frac{V - \rho\left(\frac{x-\mu_1}{\sigma_1}\right)}{\sqrt{1-\rho^2}} = W$

$$V - \rho\left(\frac{x-\mu_1}{\sigma_1}\right) = \sqrt{1-\rho^2} W$$

$$V = \rho\left(\frac{x-\mu_1}{\sigma_1}\right) + \sqrt{1-\rho^2} W$$

$$dV = \sqrt{1-\rho^2} dW$$

$$g(x) = \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - \frac{1}{2}W^2\right] \sqrt{1-\rho^2} dW$$

$$= \frac{1}{\sqrt{2\pi} \cdot \sqrt{2\pi} \sigma_1} \int_{-\infty}^{\infty} \left(e^{-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2} \cdot e^{-\frac{1}{2}W^2} \right) dW$$

$$g(x) = \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}w^2} dw \quad (1)$$

put $\frac{w^2}{2} = t$ $w^2 = 2t$
 $w = \sqrt{2t}$

$$2w dw = 2 dt$$

$$dw = \frac{dt}{w} = \frac{dt}{\sqrt{2t}}$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2} \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-t} \cdot \frac{dt}{\sqrt{2t}}$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2} \frac{1}{\sqrt{\pi}} \int_0^{\infty} t^{-\frac{1}{2}} e^{-t} dt$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2} \frac{1}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2}$$

$$g(x) = \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2}$$

the marginal density function of the variable y_1 is.

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Do yourself.

Question
 If (X, Y) has a Bivariate Normal Distribution, Then the
 Condition Distribution of X given $Y=y$ is normal
 with mean $[\mu_1 + \rho \frac{\sigma_1}{\sigma_2} (y - \mu_2)]$ and Variance $\sigma_1^2(1-\rho^2)$.
 Also, The conditional distribution of Y given $X=x$ is normal
 with mean $[\mu_2 + \rho \frac{\sigma_2}{\sigma_1} (x - \mu_1)]$ and Variance $\sigma_2^2(1-\rho^2)$

Sol. The conditional density of X for fixed value
 of Y is.

$$f(X/Y) = \frac{f(X, Y)}{h(Y)}$$

$$f(X/Y) = \frac{1}{2\pi \sigma_1 \sigma_2 \sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{X-\mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{X-\mu_1}{\sigma_1} \right) \left(\frac{Y-\mu_2}{\sigma_2} \right) + \left(\frac{Y-\mu_2}{\sigma_2} \right)^2 \right\} \right]$$

$$\div \frac{1}{\sqrt{2\pi} \sigma_2} \exp \left[-\frac{1}{2} \left(\frac{Y-\mu_2}{\sigma_2} \right)^2 \right]$$

$$= \frac{1}{\sqrt{2\pi} \sigma_1 \sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{X-\mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{X-\mu_1}{\sigma_1} \right) \left(\frac{Y-\mu_2}{\sigma_2} \right) + \left(\frac{Y-\mu_2}{\sigma_2} \right)^2 \right\} \right]$$

$$\times \exp \left[\frac{1}{2} \left(\frac{Y-\mu_2}{\sigma_2} \right)^2 \right]$$

$$= \frac{1}{\sqrt{2\pi} \sigma_1 \sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{X-\mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{X-\mu_1}{\sigma_1} \right) \left(\frac{Y-\mu_2}{\sigma_2} \right) + \left(\frac{Y-\mu_2}{\sigma_2} \right)^2 - (1-\rho^2) \left(\frac{Y-\mu_2}{\sigma_2} \right)^2 \right\} \right]$$

$$= \frac{1}{\sqrt{2\pi} \sigma_1 \sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{X-\mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{X-\mu_1}{\sigma_1} \right) \left(\frac{Y-\mu_2}{\sigma_2} \right) + \left(\frac{Y-\mu_2}{\sigma_2} \right)^2 - \left(\frac{Y-\mu_2}{\sigma_2} \right)^2 + \rho^2 \left(\frac{Y-\mu_2}{\sigma_2} \right)^2 \right\} \right]$$

$$= \frac{1}{\sqrt{2\pi} \sigma_1 \sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{X-\mu_1}{\sigma_1} \right)^2 - \rho \left(\frac{Y-\mu_2}{\sigma_2} \right) \right\}^2 \right]$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)} \left\{ \frac{\sigma_2(x-\mu_1) - \rho\sigma_1(y-\mu_2)}{\sigma_1\sigma_2} \right\}^2\right]$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)} \left\{ \sigma_1 x - \sigma_2 \mu_1 - \rho\sigma_1 y + \rho\sigma_2 \mu_2 \right\}^2\right]$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)} \left\{ \sigma_1 \left(x - \mu_1 - \rho \frac{\sigma_1}{\sigma_2} y + \rho \frac{\sigma_2}{\sigma_1} \mu_2 \right) \right\}^2\right]$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)} \left\{ \frac{\sigma_1^2}{\sigma_1^2} \left(x - \left(\mu_1 + \rho \frac{\sigma_1}{\sigma_2} y - \rho \frac{\sigma_2}{\sigma_1} \mu_2 \right) \right)^2 \right\}\right]$$

$$f(y/x) = \frac{1}{\sqrt{2\pi}\sigma_1\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)} \left\{ \frac{x - \left(\mu_1 + \rho \frac{\sigma_1}{\sigma_2} (y - \mu_2) \right)}{\sigma_1} \right\}^2\right]$$

which is a univariate normal density with mean $\mu = \mu_1 + \rho \frac{\sigma_1}{\sigma_2} (y - \mu_2)$ and Variance

$$\sigma^2 = (1 - \rho^2) \sigma_1^2$$

Now

The conditional density of Y for a fixed value of x is

$$f(y/x) = \frac{f(x, y)}{f(x)}$$

Do yourself